

Question Paper

Exam Date & Time: 27-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Software Engineering [NPT_4123]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Compare unstructured and structured programming paradigms. Using appropriate examples, analyze how program flow and readability differ between the two approaches. (5)
- A) Course Outcome: "CO1" / Cognitive Level: "Application"
- B) Demonstrate your understanding of software development by distinguishing between software as a service (SaaS) and software products and provide appropriate examples for each. (5)
- Course Outcome: "CO1" / Cognitive Level: "Analysis"
- 2) Analyze the Prototype Life Cycle Model and explain how it helps in addressing ambiguity or uncertainty in software requirements during the development process. (5)
- A) Course Outcome: "CO2" / Cognitive Level: "Analysis"
- B) Explain the concepts of functional and non-functional requirements in a software system. Using your understanding, identify 3 functional requirements and two non-functional requirement for an online shopping system. (5)
- Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 3) Analyze the characteristics of high-level and detailed designs, and determine the specific documents that should be produced upon the completion of each design stage. (5)
- A) Course Outcome: "CO3" / Cognitive Level: "Application"
- B) Analyze the concepts of coupling and cohesion, and describe how communicational and sequential cohesion are applied in software module design. (5)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- 4) An Online Shopping System enables users to log in, browse products, add selected items to their cart, make secure payments, and receive an order confirmation. The key operations of the system include: user authentication, product selection, cart management, payment processing, and order confirmation. Using the concepts of UML Component Diagrams, apply your understanding to construct a component diagram for this system. (5)
- A) Course Outcome: "CO4" / Cognitive Level: "Application"
- B) Use your understanding of object-oriented modeling to **construct** a Class for a Smart Agriculture (5)

Monitoring System that employs IoT sensors to measure soil moisture, temperature, and humidity in farmland. The system should continuously collect environmental data and analyze it to determine when irrigation is required. When the soil moisture level falls below a predefined threshold, the system must send an instant notification to the farmer's mobile device, alerting them to initiate irrigation. All sensor readings should be stored in a central database for further analysis and decision support.

Identify the main classes, their attributes, operations, and appropriate relationships

Course Outcome: "CO4" / Cognitive Level: "Application"

5) Consider the following C function named sort. (5)

/* sort takes an integer array and sorts it in ascending order */

A)

```
void sort(int a[], int n){
    int i,j;
    for(i=0;i<n-1;i++)
        for(j=i+1;j<n;j++)
            if(a[i]>a[j])
            {
                temp=a[i];
                a[i]=a[j];
                a[j]=temp;
            }
}
```

Explain the following

1. Decision coverage
2. Branch coverage

Course Outcome: "CO5" / Cognitive Level: "Analysis"

B) Analyse the requirements and design a black-box test suite for a program that takes two strings as (5)

input, identifies if the first string is a substring of the second, and displays the number of occurrences. Justify how your chosen test cases achieve comprehensive coverage of the program's expected behaviours and possible edge cases.

Course Outcome: "CO5" / Cognitive Level: "Analysis"

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Event Name	Subject
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#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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No of

#	Level	Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	5	25(50%)
4	Analysis	5	25(50%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	10	50 (100%)